

Mean Deviation

- Mean deviation calculated by any measure of central tendency is an absolute measure.
- For the purpose of comparing variation among different series, a relative mean deviation is required.
- The relative mean deviation is obtained by dividing the mean deviation by the average used for calculating mean deviation.

$$\text{Coefficient of mean deviation} = \frac{\text{Mean deviation}}{\text{Mean or Median or Mode}}$$

Mean deviation-Ungrouped data

- Calculate the average(mean, median or mode) of the series.
- Take the deviations of items from average ignoring signs and denote these deviations by $|D|$.
- Compute the total of these deviations, i.e., $\sum |D|$
- Divide this total obtained by the number of items.

Example 2

Calculate mean deviation from mean and median for the following data: 100, 150, 200, 250, 360, 490, 500, 600, 671. Also, calculate coefficients of M.D.

Solution

Mean $\bar{x} = \frac{\sum x}{n} = \frac{3321}{9} = 369$. Median = 360.

Wages	$ D = x - \bar{x}$	$ D = x - \text{median}$
100	269	260
150	219	210
200	169	160
250	119	110
360	9	0
490	121	130
500	131	140
600	231	240
671	302	311
$\sum x = 3321$	$\sum D = 1570$	$\sum D = 1561$

Solution Cont...

- Mean deviation(from Mean) $= \frac{\sum |D|}{n} = \frac{1570}{9} = 174.44$.
- Coefficient of M.D(from mean) $= \frac{M.D}{\bar{x}} = \frac{174.44}{369} = 0.4727$.
- Mean deviation(from Median) $= \frac{\sum |D|}{n} = \frac{1561}{9} = 173.44$.
- Coefficient of M.D(from Median) $= \frac{M.D}{Median} = \frac{173.44}{360} = 0.4817$.

Mean deviation-Grouped data:Discrete Series

- Calculate the average(mean, median or mode) of the series.
- Find out the deviation of the variable values from the average, ignoring signs and denote these deviations by $|D|$.
- Compute the total of these deviations, i.e., $\sum f|D|$
- Divide this total obtained by the total frequency N . i.e.,

$$M.D = \frac{\sum f|D|}{N}$$

Problem 10

Calculate Mean deviation from the following table.

x	2	4	6	8	10
f	1	4	6	4	1

Solution

x	f	fx	$ D = x - \bar{x} $	$f D $
2	1	2	4	4
4	4	16	2	8
6	6	36	0	0
8	4	32	2	8
10	1	10	4	4
$N = \sum f = 16$		$\sum fx = 96$		$\sum f D = 24$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{96}{16} = 6.$$

$$\text{Mean deviation} = \frac{\sum f|D|}{\sum f} = \frac{24}{16} = 1.5.$$

Problem 11

*Calculate mean deviation from mean and median for the following data:
Also, calculate coefficients of M.D*

<i>Height in cms</i>	<i>Number of Persons</i>
158	15
159	20
160	32
161	35
162	33
163	22
164	10
165	20
166	8

Mean deviation-Grouped data:continuous Series

- Calculate the average (mean, median or mode) of the series.
- Compute $|D| = |Mid - Average|$, where Mid is the mid point of class interval.
- Compute $\sum f|D|$
- Divide this total obtained by the total frequency N .

$$M.D = \frac{\sum f|D|}{N}$$

Problem 12

*Calculate mean deviation from mean and median for the following data:
Also, calculate coefficients of Mean deviation.*

Marks	Number of students
0-10	20
10-20	25
20-30	32
30-40	40
40-50	42
50-60	35
60-70	10
70-80	8

Solution

Here $\bar{x} = 36.5$ and Median = 37.25, $D_1 = |Mid - \bar{x}|$ and $|D_2| = |Mid - Median|$.

Class	f	Mid	f(Mid)	cf	$ D_1 $	$f(D_1)$	$ D_2 $	$f(D_2)$
0-10	20	5	100	20	31.5	630	32.25	645
10-20	25	15	375	45	21.5	537.5	22.25	556.25
20-30	32	25	800	77	11.5	368	12.25	392
30-40	40	35	1400	117	1.5	60	2.25	90
40-50	42	45	1890	159	8.5	357	7.75	325.5
50-60	35	55	1925	194	18.5	647.5	17.75	621.25
60-70	10	65	650	204	28.5	285	27.75	277.5
70-80	8	75	600	212	38.5	308	37.75	302
Total		212	7740			3193		3209.5

Solution Cont...

- To find Mean and Median.

Mean $\bar{x} = \frac{\sum fMid}{N}$, where Mid is the mid point of the class interval and N is the total frequency.

Therefore, $\bar{x} = 36.5$

- To find Median

Now, $\frac{N}{2} = \frac{212}{2} = 106$

Therefore, Median class is 30-40

$$\text{Median } Md = l + h \left(\frac{\frac{N}{2} - cf}{f} \right)$$

Here $l = 30$, $cf = 77$, $f = 40$, $h = 10$.

Hence Median $Md = 37.25$.

Solution Cont...

Mean deviation from mean

$$M.D_1 = \frac{\sum f|D_1|}{N} = \frac{3193}{212} = 15.0613$$

Coefficient of mean deviation from mean

$$= \frac{\text{Mean Deviation}}{\text{Mean}} = \frac{15.0613}{36.5} = 0.4126$$

Mean deviation from median

$$M.D_2 = \frac{\sum f|D_2|}{N} = \frac{3209.5}{212} = 15.1391$$

Coefficient of mean deviation from median

$$= \frac{\text{Mean Deviation}}{\text{Median}} = \frac{15.1391}{37.25} = 0.4064$$

Problem 13

Calculate Mean deviation from the mean and median the following table.

<i>Height(in cms)</i>	<i>158</i>	<i>159</i>	<i>160</i>	<i>161</i>	<i>162</i>	<i>163</i>	<i>164</i>	<i>165</i>	<i>166</i>
<i>No. of students</i>	<i>15</i>	<i>20</i>	<i>32</i>	<i>35</i>	<i>33</i>	<i>22</i>	<i>20</i>	<i>10</i>	<i>8</i>

Standard Deviation and Coefficient of variation

It is defined as the positive square-root of the arithmetic mean of the Square of the deviations of the given observation from their arithmetic mean. The standard deviation is denoted by σ . **Standard deviation for**

Ungrouped data

- $\sigma = \sqrt{\frac{\sum(x-\bar{x})^2}{n}}$, where $\bar{x}$ is the mean.
- $\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$, where $d = x - A$ and A is the assumed value
- Variance = σ^2

Problem 14

Calculate the standard deviation from the following data. 14, 22, 9, 15, 20, 17, 12, 11.

Solution: $\bar{x} = 15$, $\sum(x - \bar{x})^2 = 140$ and $\sigma = 4.18$.

Coefficient of variation(C.V)

The Standard deviation is an absolute measure of dispersion. It is expressed in terms of units in which the original figures are collected and stated. The standard deviation of heights of students cannot be compared with the standard deviation of weights of students, as both are expressed in different units, i.e heights in centimeter and weights in kilograms. Therefore the standard deviation must be converted into a relative measure of dispersion for the purpose of comparison. The relative measure is known as the coefficient of variation.

$$\text{Coefficient of standard deviation} = \frac{\sigma}{\bar{X}}$$

$$\text{Coefficient of variation(C.V)} = \frac{\text{standard deviation}}{\text{Mean}} \times 100 = \frac{\sigma}{\bar{X}} \times 100.$$

Standard deviation for Grouped data-Discrete Series

$$\sigma = \sqrt{\frac{\sum f(x-\bar{x})^2}{N}}, \text{ where } \bar{x} \text{ is the mean and } N \text{ is the total frequency.}$$

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2}, \text{ where } d = x - A \text{ and } A \text{ is the assumed value}$$

Problem 15

Calculate Standard deviation from the following data.

X	20	22	25	31	35	40	42	45
f	5	12	15	20	25	14	10	6

Standard deviation for Grouped data- Continuous Series

$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times c$, where $d = \frac{x-A}{c}$ and A is the assumed value, c is the interval length and N is total frequency..

Problem 16

Calculate Standard Deviation for the following data.

Marks	Number of students
0-10	6
10-20	5
20-30	8
30-40	15
40-50	7
50-60	6
60-70	3

Solution

x	Mid= m	f	$d = \frac{m-35}{10}$	fd	fd^2
0-10	5	6	-3	-18	54
10-20	15	5	-2	-10	20
20-30	25	8	-1	-8	8
30-40	35	15	0	0	0
40-50	45	7	1	7	7
50-60	55	6	2	12	24
60-70	65	3	3	9	27
Total		$N=50$		-8	140

Solution Cont...

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times c \\&= \sqrt{\frac{140}{50} - \left(\frac{-8}{50}\right)^2} \times 10 \\&= \sqrt{\frac{7000}{2500} - \frac{64}{2500}} \times 10 \\&= \sqrt{\frac{6936}{2500}} \times 10 = 1.6656 \times 10 = 16.65.\end{aligned}$$

Problem 17

Find the coefficient of variation for the above problem.

Problem 18

Prices of a particular commodity in five years in two cities are given below:

<i>Price in City A</i>	<i>Price in City B</i>
20	10
22	20
19	18
23	12
16	15

Which city has more stable prices?

Problem 19

Calculate the mean and standard deviation for the following table giving the age distribution of 542 members. table.

Age(in years)	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of Members	3	61	132	153	140	51	2

Solution: $\bar{x} = 54.72$ years and $\sigma = 11.88$ years.

Moments

Moments can be defined as the arithmetic mean of various powers of deviations taken from the mean of a distribution. These moments are known as **central moments**.

- Individual series (un grouped data):

$$\mu_r = \frac{\sum (x - \bar{x})^r}{n}$$

- Discrete series:

$$\mu_r = \frac{\sum f(x - \bar{x})^r}{N}$$

- Continuous Series:

$$\mu_r = \frac{\sum f(mid - \bar{x})^r}{N},$$

where *mid* is the mid point of each class.

- Note that $\mu_1 = 0$ and $\mu_2 = \sigma^2$.

Raw moments

If the mean is a fractional value, then it becomes a difficult task to work out the moments. In such cases, we can calculate moments about a working origin and then change it into moments about the actual mean. The moments about an origin are known as **raw moments** or **moment about origin**.

Raw Moments

- Individual series (un grouped data):

$$\mu'_r = \frac{\sum d_1^r}{n},$$

where $d_1 = (x - A)$ and A is any origin.

- Discrete series:

$$\mu'_r = \frac{\sum fd_2^r}{N} \times C^r,$$

where $d_2 = \frac{x-A}{C}$ and C is the common point.

- Continuous Series

$$\mu'_r = \frac{\sum fd_3^r}{N} \times C^r,$$

where $d_3 = \frac{\text{mid}-A}{C}$ and C is the class interval width.

Relation between moments about arithmetic mean and moments about an origin are given below.

$$\mu_1 = \mu'_1 - \mu'_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

Problem 20

From the data given below, calculate the first four raw and central moments

X	f
30-33	2
33-36	4
36-39	26
39-42	47
42-45	15
45-48	6

Problem 21

Calculate the first moments of the following table about the mean and hence find β_1 and β_2

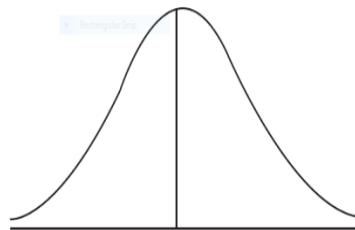
x	0	1	2	3	4	5	6	7	8
f	1	8	28	56	70	56	28	8	1

ANS: $\mu_1 = 0, \mu_2 = 2, \mu_3 = 0, \mu_4 = 11, \beta_1 = 0$ and $\beta_2 = 2.75$.

Skewness

Skewness means 'lack of symmetry'. We study skewness to have an idea about the shape of the curve which we can draw with the help of the given data. If in a distribution $\text{mean} = \text{median} = \text{mode}$, then that distribution is known as *symmetrical distribution*. If in a distribution $\text{mean} \neq \text{median} \neq \text{mode}$, then it is *not a symmetrical* distribution and it is called a skewed distribution and such a distribution could either be *positively skewed or negatively skewed*.

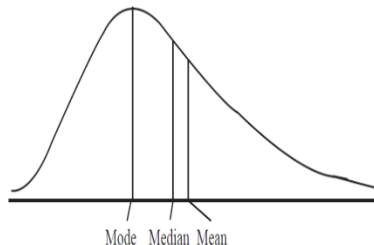
Symmetrical distribution



Mean = Median = Mode

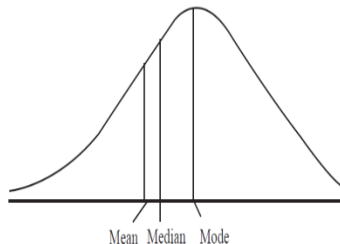
It is clear from the above diagram that in a symmetrical distribution the values of mean, median and mode coincide. The spread of the frequencies is the same on both sides of the center point of the curve.

Positively skewed distribution



It is clear from the above diagram, in a positively skewed distribution, the value of the mean is maximum and that of the mode is least, the median lies in between the two. In the positively skewed distribution the frequencies are spread out over a greater range of values on the right hand side than they are on the left hand side.

Negatively skewed distribution



It is clear from the above diagram, in a negatively skewed distribution, the value of the mode is maximum and that of the mean is least. The median lies in between the two. In the negatively skewed distribution the frequencies are spread out over a greater range of values on the left hand side than they are on the right hand side.

Measures of skewness

- Karl-Pearason's coefficient of skewness
- Bowley's coefficient of skewness
- Measure of skewness based on moments

- Karl Pearson's Coefficient Skewness =

$$\frac{Mean - Mode}{\sigma}$$

- Bowley's Coefficient of skewness=

$$\frac{Q_3 + Q_1 - 2Median}{Q_3 - Q_1}$$

- The measure of skewness based on moments is denoted by β_1 and is given by:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Problem 22

Calculate Kerl Pearson's Coefficient of Skewness for the following data.
25, 15, 23, 40, 27, 25, 23, 25, 20.

ANS: $\bar{x} = 24.75$, $\sigma = 6.3$, Mode is 25 and coefficient is -0.03.

Problem 23

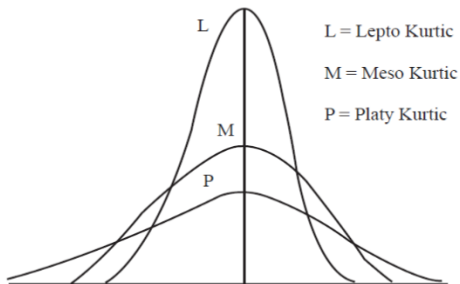
Find the Kerl Pearson's coefficient of skewness for the following data given below.

x	3	4	5	6	7	8	9	10
f	7	10	14	35	102	136	43	8

ANS: Mean=7.35, Mode=8, $\sigma = 1.33$ and coefficient =-0.49

Measure of kurtosis

Measure of kurtosis denote the shape of top of a frequency curve. Measure of kurtosis tell us the extent to which a distribution is more peaked or more flat topped than the normal curve, which is symmetrical and bellshaped, is designated as **Mesokurtic**. If a curve is relatively more narrow and peaked at the top, it is designated as **Leptokurtic**. If the frequency curve is more flat than normal curve, it is designated as **platykurtic**.



Measure of Kurtosis

The measure of kurtosis of a frequency distribution based moments is denoted by β_2 and is given by

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

- If $\beta_2 = 3$, the distribution is said to be normal and the curve is **mesokurtic**.
- If $\beta_2 > 3$, the distribution is said to be more peaked and the curve is **leptokurtic**.
- If $\beta_2 < 3$, the distribution is said to be flat topped and the curve is **platykurtic**.